

math-blurbs

self-adjoint

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This project is hosted on GitHub: <https://github.com/self-adjoint/math-blurbs>

0 Introduction

0.1 The QED Symbol

QED stands for *quod erat demonstrandum* (Latin for “that which was to be demonstrated”), and is used to signify the end of a mathematical proof. Most mathematicians use either a white square $\square$ or a black square $\blacksquare$ for this. **Paul J., Sally, Jr.** (1933 – 2013) famously used a sketch of himself in an eyepatch¹ as the QED symbol in his books.

Theorem 0.1. For any group of people, there are at least two people with the same number of friends.

Proof. Suppose there are n people in the group. Each person can have any number of friends from 0 to $n - 1$ (so there are n possible values). However, there cannot be both a person with 0 friends (friends with no one) and a person with $n - 1$ friends (friends with everyone). Thus, there must be at least two people with the same number of friends. 😎

The QED symbol I have chosen to use is the **Smiling Face with Sunglasses** emoji². This version has been used by **WhatsApp** since 2017.

Theorem 0.2 (Cool thing). This is a theorem

Theorem 0.2

¹He was legally blind in his left eye, and nicknamed “The Math Pirate” and “Professor Pirate” by his students.

²Not to be confused with an **emoticon**, which refers to things like :) and :(and were mainly used before emojis were introduced into **Unicode 6.0**.

1 Limit inferior and limit superior

Definition 1.1. Suppose $(a_n)_{n=1}^{\infty}$ is a sequence in $\mathbb{R}$. The **limit inferior** and **limit superior** of $(a_n)_{n=1}^{\infty}$ respectively are given by:

$$\liminf_{n \rightarrow \infty} a_n = \sup_{n \in \mathbb{N}} \inf_{k \geq n} a_k \qquad \limsup_{n \rightarrow \infty} a_n = \inf_{n \in \mathbb{N}} \sup_{k \geq n} a_k$$

Note that $\liminf_{n \rightarrow \infty} a_n$ and $\limsup_{n \rightarrow \infty} a_n$ are allowed to be $\pm\infty$.

1.1

Suppose $(a_n)_{n=1}^{\infty}$ is a sequence in $\mathbb{R}$ and $x \in \mathbb{R}$. Prove that:

1. $x < \liminf_{n \rightarrow \infty} a_n$ if and only if $x < a_n$ for all but finitely many $n \in \mathbb{N}$.
2. $x < \limsup_{n \rightarrow \infty} a_n$ if and only if $x < a_n$ for infinitely many $n \in \mathbb{N}$.

Do these statements hold if we replace the strict inequalities with non-strict inequalities? Prove or give a counterexample for each one.

1.2

Suppose $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ are sequences in $\mathbb{R}$. Prove that:

$$\liminf_{n \rightarrow \infty} a_n + \liminf_{n \rightarrow \infty} b_n \leq \liminf_{n \rightarrow \infty} (a_n + b_n) \leq \liminf_{n \rightarrow \infty} a_n + \limsup_{n \rightarrow \infty} b_n \leq \limsup_{n \rightarrow \infty} (a_n + b_n) \leq \limsup_{n \rightarrow \infty} a_n + \limsup_{n \rightarrow \infty} b_n$$

For each inequality, give an example of sequences $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ where it is strict. Can you find a single example of $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ where *all four* inequalities are strict?

Definition 1.2. Suppose $(A_n)_{n=1}^{\infty}$ is a sequence of subsets of $\mathbb{R}$. The **limit inferior** and **limit superior** of $(A_n)_{n=1}^{\infty}$ respectively are given by:

$$\liminf_{n \rightarrow \infty} A_n = \bigcup_{n=1}^{\infty} \bigcap_{k \geq n} A_k \qquad \limsup_{n \rightarrow \infty} A_n = \bigcap_{n=1}^{\infty} \bigcup_{k \geq n} A_k$$

1.3

Suppose $(A_n)_{n=1}^{\infty}$ is a sequence of subsets of $\mathbb{R}$ and $x \in \mathbb{R}$. Prove that:

1. $x \in \liminf_{n \rightarrow \infty} A_n$ if and only if $x \in A_n$ for all but finitely many $n \in \mathbb{N}$.
2. $x \in \limsup_{n \rightarrow \infty} A_n$ if and only if $x \in A_n$ for infinitely many $n \in \mathbb{N}$.

1.4

State and prove an analog of **Exercise 1.2** for sequences of subsets of $\mathbb{R}$.

Definition 1.3. A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is **additive** if $f(x+y) = f(x) + f(y)$ for all $x, y \in \mathbb{R}$.

Remarks.

1. The equation $f(x+y) = f(x) + f(y)$ is known as the *Cauchy functional equation*.
2. It follows by replacing x with $x-y$ that $f(x-y) = f(x) - f(y)$ for all $x, y \in \mathbb{R}$. It then follows by substituting $x=y$ that $f(0) = 0$.

Lemma 1.4. Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is additive. Then $f(cx) = cf(x)$ for all $c \in \mathbb{Q}$ and all $x \in \mathbb{R}$.

Proof. Suppose $x \in \mathbb{R}$. We first show that $f(nx) = nf(x)$ for all $n \in \mathbb{Z}$. Clearly this holds for $n = 1$ as both sides reduce to $f(x)$. Suppose the equation holds for some $n \in \mathbb{Z}$. Then we have:

$$\begin{aligned} f((n+1)x) &= f(nx+x) = f(nx) + f(x) = nf(x) + f(x) = (n+1)f(x) \\ f((n-1)x) &= f(nx-x) = f(nx) - f(x) = nf(x) - f(x) = (n-1)f(x) \end{aligned}$$

Thus the equation holds for $n+1$ and $n-1$, and so it holds for all $n \in \mathbb{Z}$.

Now suppose $c \in \mathbb{Q}$. Then $c = \frac{p}{q}$ for some $p, q \in \mathbb{Z}$, $q \neq 0$. By what we have just shown, we have $f(px) = pf(x)$ and $f(qy) = qf(y)$ for all $x, y \in \mathbb{R}$. Substituting $y = \frac{x}{q}$ yields $f(x) = qf(\frac{x}{q})$, i.e. $f(\frac{x}{q}) = \frac{1}{q}f(x)$, and so $f(\frac{p}{q}x) = \frac{p}{q}f(x)$.

Thus $f(cx) = cf(x)$ for all $c \in \mathbb{Q}$ and all $x \in \mathbb{R}$. 🧐

Proposition 1.5. If $f : \mathbb{R} \rightarrow \mathbb{R}$ is additive and Lebesgue measurable, then f is continuous at 0.

Proof. Suppose $U \subseteq \mathbb{R}$ is an open set containing 0 and $V \subseteq U$ is an open set such that $\{x - y \mid x, y \in V\} \subseteq U$. Also suppose $(q_n)_{n=1}^\infty$ is an enumeration of $\mathbb{Q}$. Then we have $\bigcup_{n=1}^\infty (q_n + V) = \mathbb{R}$, and so $\bigcup_{n=1}^\infty f^{-1}(q_n + V) = \mathbb{R}$ as well. For each $n \in \mathbb{N}$, define $W_n = f^{-1}(q_n + V)$. Then we have $\bigcup_{n=1}^\infty W_n = \mathbb{R}$, so $\mu(W_n) > 0$ for at least one $n \in \mathbb{N}$. By Sierpinski's theorem on sets of differences, the set $\{x - y \mid x, y \in W_n\}$ contains $(-\varepsilon, \varepsilon)$ for some $\varepsilon > 0$. Since $\{x - y \mid x, y \in W_n\} \subseteq f^{-1}(\{x - y \mid x, y \in V\}) \subseteq f^{-1}(U)$, it follows that $f^{-1}(U)$ contains $(-\varepsilon, \varepsilon)$. Thus f is continuous at 0. 🧐

Lemma 1.6. If $f : \mathbb{R} \rightarrow \mathbb{R}$ is additive and continuous at one point, then it is continuous on $\mathbb{R}$.

Proof. Suppose f is continuous at some $a \in \mathbb{R}$. Then we have $\lim_{x \rightarrow a} f(x) = f(a)$. For any $b \in \mathbb{R}$, we have:

$$\lim_{x \rightarrow b} f(x) = \lim_{x \rightarrow a} f(x - a + b) = \lim_{x \rightarrow a} (f(x) - f(a) + f(b)) = \lim_{x \rightarrow a} f(x) - f(a) + f(b) = f(b)$$

Thus f is continuous at every $b \in \mathbb{R}$. 🧐

Proposition 1.7. Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is additive and continuous. Then there is a constant $a \in \mathbb{R}$ such that $f(x) = ax$ for all $x \in \mathbb{R}$.

Proof. Define $a = f(1)$. By [Lemma 1.4](#), we have $f(cy) = cf(y)$ for all $c \in \mathbb{Q}$ and all $y \in \mathbb{R}$. Setting $y = 1$ yields $f(c) = cf(1) = ac$. Suppose $x \in \mathbb{R}$ and (q_n) is a sequence in $\mathbb{Q}$ that converges to x . Since f is continuous, the sequence $(f(q_n))$ converges to $f(x)$. This yields $f(x) = \lim_{n \rightarrow \infty} f(q_n) = \lim_{n \rightarrow \infty} aq_n = a \lim_{n \rightarrow \infty} q_n = ax$. Thus $f(x) = ax$ for all $x \in \mathbb{R}$. 🧐

Theorem 1.8. Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is additive and Lebesgue measurable. Then there is a constant $a \in \mathbb{R}$ such that $f(x) = ax$ for all $x \in \mathbb{R}$.

Proof. By [Proposition 1.5](#), f is continuous at 0, so by [Lemma 1.6](#), it is continuous on $\mathbb{R}$, and so by [Proposition 1.7](#), we have $f(x) = ax$ for all $x \in \mathbb{R}$, where $a = f(1)$ is a constant. 🧐

Corollary 1.9. Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is Lebesgue measurable and $f(x + y) = f(x)f(y)$ for all $x, y \in \mathbb{R}$. Then either $f \equiv 0$ or there is a constant $a > 0$ such that $f(x) = a^x$ for all $x \in \mathbb{R}$.

Proof. Clearly $f \equiv 0$ satisfies the above conditions. Note that if $f(x) = 0$ for some $x \in \mathbb{R}$, then $f(x + y) = 0f(y) = 0$ for all $y \in \mathbb{R}$, and so $f \equiv 0$. Suppose $f \not\equiv 0$. Then $f(x) \neq 0$ for all $x \in \mathbb{R}$. Moreover, we have $f(x) > 0$ for all $x \in \mathbb{R}$, since $f(x) = f(\frac{x}{2})^2$. Define $g : \mathbb{R} \rightarrow \mathbb{R}$ by $g(x) = \ln(f(x))$. This is well-defined as $f(x) > 0$ for all $x \in \mathbb{R}$. We also have $g(x + y) = \ln(f(x + y)) = \ln(f(x)f(y)) = \ln(f(x)) + \ln(f(y)) = g(x) + g(y)$. Thus g is additive. It is also Lebesgue measurable as $g = \ln(f)$ and $\ln$ is continuous. By [Theorem 1.8](#), we have $g(x) = bx$ for all $x \in \mathbb{R}$, where $b = g(1)$ is a constant. Thus $f(x) = e^{bx} = a^x$ for all $x \in \mathbb{R}$, where $a = e^b$. 🧐

Since $\mathbb{R}$ is a vector space over $\mathbb{Q}$, it has a basis. In other words, there is a set $\mathcal{B} \subseteq \mathbb{R}$ such that every $x \in \mathbb{R}$ can be expressed uniquely as $x = q_1x_1 + q_2x_2 + \dots + q_nx_n$, where $x_1, x_2, \dots, x_n \in \mathcal{B}$ and $q_1, q_2, \dots, q_n \in \mathbb{Q}$. We now assign arbitrary values to $f(b)$ for every $b \in \mathcal{B}$. Since $\mathcal{B}$ is linearly independent over $\mathbb{Q}$, the value of $f(b)$ for any given $b \in \mathcal{B}$ is not affected by the values of f at other points in $\mathcal{B}$, i.e. whatever values f takes at points in $\mathcal{B}$, they will not contradict the equation $f(x + y) = f(x) + f(y)$. Also, since $\mathcal{B}$ spans $\mathbb{R}$, the values of f at points in $\mathcal{B}$ uniquely determine its values at every point in $\mathbb{R}$.

Definition 1.10. Suppose $I \subseteq \mathbb{R}$ is an interval. A function $f : I \rightarrow \mathbb{R}$ is **midpoint convex** (or *Jensen convex* or *midconvex*) on I if $f\left(\frac{x+y}{2}\right) \leq \frac{f(x)+f(y)}{2}$ for all $x, y \in I$.

Remark. It follows directly (by setting $\lambda = \frac{1}{2}$ in the definition of a convex function) that every convex function is midpoint convex.

Theorem 1.11. If $f : I \rightarrow \mathbb{R}$ is midpoint convex and continuous, then f is convex.

Proof. Since f is midpoint convex, for any $x, y \in I$, we have $f\left(\frac{x+y}{2}\right) \leq \frac{f(x)+f(y)}{2}$. Since I is an interval, we also have

$\frac{x+y}{2} \in I$. This implies:

$$\begin{aligned} f\left(\frac{3}{4}x + \frac{1}{4}y\right) &= f\left(\frac{1}{2}\left(\frac{x+y}{2} + y\right)\right) \leq \frac{1}{2}\left(f\left(\frac{x+y}{2}\right) + f(y)\right) \leq \frac{1}{2}\left(\frac{f(x)+f(y)}{2} + f(y)\right) = \frac{3}{4}f(x) + \frac{1}{4}f(y) \\ f\left(\frac{1}{4}x + \frac{3}{4}y\right) &= f\left(\frac{1}{2}\left(x + \frac{x+y}{2}\right)\right) \leq \frac{1}{2}\left(f(x) + f\left(\frac{x+y}{2}\right)\right) \leq \frac{1}{2}\left(f(x) + \frac{f(x)+f(y)}{2}\right) = \frac{1}{3}f(x) + \frac{1}{4}f(y) \end{aligned}$$

Repeating this process, we have for all $n \in \mathbb{N}$ and all $m \in \{0, 1, \dots, 2^n\}$:

$$f\left(\frac{m}{2^n}x + \left(1 - \frac{m}{2^n}\right)y\right) \leq \frac{m}{2^n}f(x) + \left(1 - \frac{m}{2^n}\right)f(y)$$

Now suppose $x, y \in I$ and $\lambda \in [0, 1]$. Then there is a sequence (q_n) of rational numbers in $[0, 1]$ with denominators 2^n that converges to λ , e.g. $q_n = \frac{\lfloor 2^n \lambda \rfloor}{2^n}$. For each $n \in \mathbb{N}$, we have $f(q_n x + (1 - q_n)y) \leq q_n f(x) + (1 - q_n)f(y)$. As $n \rightarrow \infty$, the right side tends to $\lambda f(x) + (1 - \lambda)f(y)$, and since f is continuous, the left side tends to $f(\lambda x + (1 - \lambda)y)$. Thus $f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$ for all $x, y \in I$ and all $\lambda \in [0, 1]$, and so f is convex. 🧐

Proof. Suppose $f : I \rightarrow \mathbb{R}$ is midpoint convex and discontinuous at some $x \in I^\circ$. Then there is a sequence (x_n) in I such that $\lim_{n \rightarrow \infty} x_n = x$ but $\lim_{n \rightarrow \infty} f(x_n) \neq f(x)$. 🧐

Theorem 1.12. If $f : I \rightarrow \mathbb{R}$ is midpoint convex and bounded on some open subinterval $(a, b) \subseteq I$, then f is continuous (and thus convex).

Theorem 1.13 (Sierpiński's Theorem on Convex Functions). If $f : I \rightarrow \mathbb{R}$ is midpoint convex and Lebesgue measurable, then f is convex.

Theorem 1.14 (Ostrowski's Theorem on Convex Functions). If $f : I \rightarrow \mathbb{R}$ is midpoint convex and bounded on a Lebesgue measurable set with positive measure, then f is convex.

Proof. Since $E \in \mathcal{L}(\mathbb{R})$ and $\mu(E) > 0$, there is an open set $G \subseteq I$, $G \supseteq E$ such that $\mu(E) \leq \mu(G) < \frac{4}{3}\mu(E)$. Since G is open, it can be expressed as a countable union of disjoint open intervals $G = \bigcup_{n=1}^{\infty} I_n$. Thus we have $\sum_{n=1}^{\infty} \mu(E \cap I_n) \leq \sum_{n=1}^{\infty} \mu(I_n) < \frac{4}{3}\mu(E \cap I_n)$. Thus there is at least one $n \in \mathbb{N}$ such that $\mu(E \cap I_n) \leq \mu(I_n) < \frac{4}{3}\mu(E \cap I_n)$. Define $J = E \cap I_n$. Since f is bounded on E , there exists $M > 0$ such that $|f| \leq M$ on E . Suppose f is not convex. Then for any $x, y \in J$, there exists $\lambda \in [0, 1]$ such that $f(\lambda x + (1 - \lambda)y) > \lambda f(x) + (1 - \lambda)f(y)$. **NOW WHAT?** 🧐

If f is convex, then f is differentiable a.e. and its derivative is non-decreasing. Thus f is twice differentiable a.e.

2 Induction

Theorem 2.1. For every $n \in \mathbb{N}$, $\cos(nx)$ and $\frac{\sin(nx)}{\sin(x)}$ can be expressed as polynomials in $\cos(x)$.

Define the

$P(n)$: $\cos(nx)$ is a polynomial in $\cos(x)$.

$Q(n)$: $\frac{\sin(nx)}{\sin(x)}$ is a polynomial in $\cos(x)$.

Proof. Suppose $P(k)$ and $Q(k)$ are true for some $k \in \mathbb{N}$. We will show that $P(k+1)$ and $Q(k+1)$ are true.

$$\begin{aligned} \cos((k+1)x) &= \cos(x + kx) \\ &= \cos(x)\cos(kx) - \sin(x)\sin(kx) \\ &= \cos(x)\cos(kx) - \sin(x)^2 \frac{\sin(kx)}{\sin(x)} \\ &= \cos(x)\cos(kx) - (1 - \cos(x)^2) \frac{\sin(kx)}{\sin(x)} \end{aligned}$$

By the inductive hypothesis, $\cos(kx)$ and $\frac{\sin(kx)}{\sin(x)}$ are polynomials in $\cos(x)$. Thus $\cos((k+1)x)$ is a polynomial in $\cos(x)$.

Similarly, we have:

$$\begin{aligned}\frac{\sin((k+1)x)}{\sin(x)} &= \frac{\sin(x+kx)}{\sin(x)} \\ &= \frac{\sin(x)\cos(kx) + \cos(x)\sin(kx)}{\sin(x)} \\ &= \cos(kx) + \cos(x)\frac{\sin(kx)}{\sin(x)}\end{aligned}$$

By the inductive hypothesis, $\cos(kx)$ and $\frac{\sin(kx)}{\sin(x)}$ are polynomials in $\cos(x)$. Thus $\frac{\sin((k+1)x)}{\sin(x)}$ is a polynomial in $\cos(x)$. 🧐

Proof.

$$\cos((k+2)x) = 2\cos(x)\cos((k+1)x) - \cos(kx) \quad (1)$$



Theorem 2.2. For all $n \in \mathbb{Z}$, we have:

$$\begin{aligned}\cos(nx) &= 2\cos(x)\cos((n-1)x) - \cos((n-2)x) \\ \sin(nx) &= 2\cos(x)\sin((n-1)x) - \sin((n-2)x)\end{aligned}$$

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```
from collections.abc import Iterator

# This is an example with math. The  $n$ th Fibonacci number is  $\frac{\phi^n - (1-\phi)^n}{\sqrt{5}}$ , where
↪  $\phi = \frac{1+\sqrt{5}}{2}$  is the golden ratio.
class Math:
    @staticmethod
    def fib(n: int) -> Iterator[int]:
        """Fibonacci series up to n."""
        a, b = 0, 1
        while a < n:
            yield a
            a, b = b, a + b

result = sum(Math.fib(42))
print(f"The answer is {result}")
```

title

some text

title

some text

title

some text

This is **NOT** the same. Oh, but **NOT**. This is **NOT** * the same. Oh, but **NOT** *.

2.1

The exercise.

Solution

The solution.

```

from collections.abc import Iterator

# This is an example
class Math:
    @staticmethod
    def fib(n: int) -> Iterator[int]:
        """Fibonacci series up to n."""
        a, b = 0, 1
        while a < n:
            yield a
            a, b = b, a + b

result = sum(Math.fib(42))
print(f"The answer is {result}")

```

Now another style:

```
print('Did this work?')
```

Also, inline code looks like this.

```

def pascal_triangle(n):
    row = (1,)
    yield row
    for _ in range(n):
        row = (1,) + tuple(map(sum, pairwise(row))) + (1,)
        yield row

```

For example, `list(pascal_triangle(5))` returns:

```
[(1,), (1, 1), (1, 2, 1), (1, 3, 3, 1), (1, 4, 6, 4, 1), (1, 5, 10, 10, 5, 1)]
```

$$\binom{6}{4} \binom{5}{3}$$

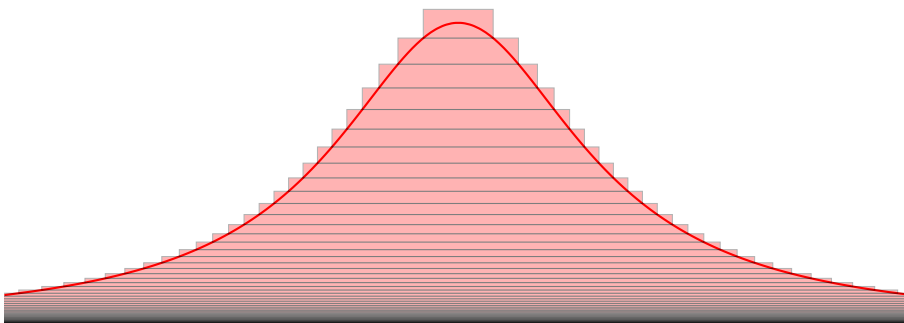
$$\lim_{k \rightarrow \infty} f(k) = \lim_{k \rightarrow \infty} f(k) = 0$$

$$\lim_{k \rightarrow \infty} f(k) = \lim_{k \rightarrow \infty} f(k) = 0$$

This appears in <https://mathoverflow.net/a/296540>

Theorem 2.3. Suppose $(X, \mathcal{A}, \mu)$ is a measure space and $f : X \rightarrow [0, \infty]$ is measurable. Then:

$$\int f d\mu = \lim_{c \rightarrow 1^+} \sum_{n=-\infty}^{\infty} c^n \mu(f^{-1}([c^n, c^{n+1})))$$



Suppose Define $f_m(x) = f(x)^{\lceil \log_c(f(x)) \rceil}$